

Engineering AI-Powered Accessibility Infrastructure for Blind and Low-Vision Users: A Case Study of YouDescribeX

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Abstract—This paper documents engineering work performed on YouDescribeX, an AI-powered accessibility platform for blind and low-vision (BLV) users, during a Research Software Engineer role at the Smith-Kettlewell Eye Research Institute in collaboration with San Francisco State University. The work spanned web, iOS, and Android surfaces and covered backend AI inference integration, an in-video question-answering feature (Infobot) built on temporal grounding, and accessibility-compliance engineering validated directly with BLV users. We report the platform, feature, and usability outcomes as documented in the author’s professional record, and discuss the engineering practices – prompt tuning for audio-description relevance, benchmark-driven retrieval evaluation, and direct usability testing with assistive-technology users – that produced them.

Index Terms—accessibility engineering, assistive technology, audio description, AI inference systems, WCAG compliance, usability testing

I. INTRODUCTION

Blind and low-vision users depend on audio description to access visual media, but professionally produced audio description is expensive to author and scales poorly to the volume of video content published daily. YouDescribeX is a platform that pairs AI-generated audio description and in-video question answering with human oversight, aiming to make video content substantially more accessible without requiring a human describer for every asset. Transforming such a system from a research prototype into a production-ready platform used by real BLV users requires engineering work across three concerns simultaneously: the AI inference pipeline that generates descriptions, the client surfaces (web, iOS, Android) that deliver them, and the accessibility compliance work that ensures the delivery mechanism itself does not introduce new barriers.

II. ROLE AND SCOPE

As a Research Software Engineer on the YouDescribeX project, the author’s scope covered three areas: (1) backend integration of AI inference for audio-description generation, (2) client-facing engineering across web, iOS, and Android, and (3) accessibility validation directly with the platform’s target user population.

III. AI INFERENCE INTEGRATION AND PROMPT TUNING

The audio-description generation path required integrating a backend AI inference service such that generated descriptions were relevant to the actual on-screen content and useful to a BLV listener relying solely on audio. Prompt tuning on this pipeline improved description relevance by 18%, a metric obtained by iterating on the instructions given to the underlying generation model against a held-out set of video content and evaluating description quality against that content. This kind of prompt-level iteration is a common and cost-effective lever in production AI systems: it improves output quality without requiring a change to the underlying model or a retraining cycle, and it can be iterated quickly against real usage data.

IV. INFOBOT: IN-VIDEO QUESTION ANSWERING WITH TEMPORAL GROUNDING

A central feature built during this engagement was Infobot, an in-video question-answering capability that allows a user to ask a question about video content and receive an answer grounded in the specific temporal segment of the video relevant to that question, rather than a generic summary of the entire asset. Infobot was evaluated against a 300-question benchmark and achieved an 86% top-3 retrieval rate, meaning that for 86% of benchmark questions, the correct grounding segment appeared within the model’s top three retrieved candidates. Top- k retrieval accuracy is a standard evaluation protocol for retrieval-augmented systems where the end system may re-rank or otherwise use multiple candidates rather than committing to a single top-1 answer, making top-3 an appropriate and conservative choice of k for a user-facing question-answering feature.

V. CROSS-PLATFORM DELIVERY AND RELIABILITY

Because BLV users access video content across devices, the platform’s audio-description and Infobot features needed to be delivered consistently on web, iOS, and Android. The author led development efforts spanning all three surfaces, integrating backend AI inference services and scalable REST APIs while preserving backward compatibility across platform releases and maintaining a crash rate below 1% across all platforms. A sub-1% crash rate across three independently engineered

client surfaces indicates that the shared backend contract and the client-specific integration work were both stable under real usage, which is a meaningful reliability bar for a platform whose users depend on it as their primary access mechanism to video content.

VI. ACCESSIBILITY VALIDATION WITH BLV USERS

Accessibility engineering claims are only meaningful when validated with the population the system is built for. The author’s work led to 100% keyboard navigation coverage and AA+ contrast compliance under NVDA and VoiceOver screen-reader testing, and usability sessions were conducted directly with 25 BLV users, reducing task completion time for common actions by 28%. This combination of automated assistive-technology testing (NVDA on the web, VoiceOver on iOS) and direct usability sessions with the target population reflects a two-tier accessibility validation strategy: automated screen-reader compatibility catches structural and semantic markup issues, while human usability testing surfaces friction that automated testing alone cannot detect, such as unclear audio-description phrasing or unintuitive navigation patterns for a non-visual user.

VII. PLATFORM OUTCOMES

Across this engagement, the platform reached 40+ pilot BLV users, with page load performance improving by 15% and 30-day user retention increasing by 20% relative to the platform’s prior state. These outcomes are reported as measured platform metrics rather than benchmarked against an external baseline, and represent the combined effect of the AI inference, cross-platform delivery, and accessibility work described above rather than the isolated contribution of any single change.

VIII. DISCUSSION

This engagement illustrates a pattern common to applied AI-for-accessibility work: the AI component (audio-description generation, in-video question answering) is necessary but not sufficient. A description that is contextually accurate but delivered through a client with poor screen-reader support, or a feature that works on one platform but not another, fails the user regardless of the underlying model’s quality. The 1% cross-platform crash-rate bar and the direct usability testing with 25 BLV users reflect an engineering practice that treats delivery reliability and accessibility validation as first-class requirements alongside AI model quality, rather than as an afterthought layered on top of a working AI pipeline.

IX. LIMITATIONS

The metrics reported in this paper are drawn from the author’s own professional record of the engagement rather than from a published, peer-reviewed evaluation of YouDescribeX, and specific methodological details of the 300-question Infobot benchmark (e.g., how questions were sourced, how ground-truth grounding segments were established) are not part of the author’s documented record and are therefore not claimed

here. Readers seeking a rigorous, independently verifiable evaluation of YouDescribeX should consult SKERI/SFSU’s own publications on the platform, if and when available.

X. CONCLUSION

This case study documents applied engineering work at the intersection of AI inference, cross-platform software delivery, and accessibility validation, performed on the YouDescribeX platform for blind and low-vision users. The reported outcomes – an 18% relevance improvement from prompt tuning, an 86% top-3 retrieval rate for in-video question answering, sub-1% cross-platform crash rates, and a 28% task-time reduction validated with 25 BLV users – reflect a development approach that treats AI quality, delivery reliability, and human-validated accessibility as jointly necessary, rather than treating any one of them as sufficient on its own.